Parameter-Free Spectral–Spatial Optimization Algorithm for Semiblind Hyperspectral and Multispectral Image Fusion

Jian Long, Jialin Gui, Yuanxi Peng, Yibing Zhan, *Member, IEEE*, Jun Li, Quan Zhang, and Yulei Tian

***Abstract*— Semiblind fusion of hyperspectral images (HSIs) and multispectral images (MSIs) is a critical technique for generating high-resolution HSIs (HR-HSIs) without the need for complex point spread function (PSF) estimation. Despite the broad application potential of semiblind fusion algorithms, existing methods face three major challenges. First, as imaging technology advances, the spatial resolution gap between HSIs and MSIs continues to widen, making high-magnification fusion increasingly urgent. Second, especially for deep learning-based methods, existing algorithms require meticulous hyperparameter tuning to enhance fusion quality. Third, the complex fusion process impedes the speed of image fusion. To address these challenges, we propose a parameter-free spectral–spatial optimization algorithm specifically designed to handle high-magnification differences in the semiblind fusion of HSI and MSI. This method enables fast fusion using simple matrix operations and consists of three main steps: 1) rapidly computing an initial solution using the Moore–Penrose inverse of the spatial response function (SRF); 2) constructing spectral errors from the initial solution to effectively extract spectral information; and 3) reconstructing spatial details using MSI to form spatial errors, thereby accurately reconstructing HR-HSI for high-quality fusion. Comparative experiments on two simulated and three real datasets with state-of-the-art algorithms demonstrate that our proposed semiblind fusion method not only achieves 64× high-magnification fusion but also reduces the computation time to just 23.9%–49.5% of that required by the fastest competing methods. The code is available at https://github.com/Long-ji/PFSSOA.**

***Index Terms*— High magnification, hyperspectral image (HSI) super-resolution, semiblind image fusion.**

I. Introduction

HYPERSPECTRAL imaging technology possesses the ability to capture rich spectral information of objects across hundreds or even thousands of narrow bands with nanometer-level resolution. This capability has made hyperspectral imaging widely applicable in areas such as remote sensing [1], product monitoring [2], precision agriculture [3], environmental monitoring [4], medical imaging [5], and defense [6]. However, its relatively low spatial resolution often limits further application of this technology. To address this limitation, increasing research is focused on image fusion techniques that fuse high-resolution multispectral images (HR-MSIs) with low-resolution hyperspectral images (LR-HSI) of the same scene.

Within the field of image fusion, methods can be classified into "blind," "semiblind," and "nonblind" fusion [7], depending on the assumptions made regarding the known PSF and spectral response function (SRF). "Nonblind" fusion refers to a fusion algorithm that leverages both PSF and SRF information, and "semiblind" fusion relies solely on SRF data, while "blind" fusion operates without access to either PSF or SRF. The PSF requires measurement and estimation and cannot be directly obtained from device parameters, while the SRF is an important parameter in imaging system design and can typically be obtained directly from the device manufacturer. The methods proposed by Dian et al. [8] (NSTF-SMBF) and Kanatsoulis et al. [9] (Blind STEREO) are semiblind fusion techniques that are robust and suitable for a variety of scenarios. Long et al.'s method [7] (NFF) further improves fusion quality compared to the previous two approaches. However, these semiblind algorithms are not specifically tailored for high-magnification fusion, leading to certain limitations in high-magnification contexts. Therefore, advancing high-magnification fusion techniques under semiblind conditions is of great importance.

The super-resolution reconstruction of traditional images with a magnification factor of $\geq 32\times$ or greater is referred to as high-magnification super-resolution [10]. Similarly, this article introduces the term "high-magnification fusion." However, due to the drastic reduction in the information content of LR-HSI under high magnification, a single LR-HSI pixel may correspond to thousands of HR-MSI pixels, which significantly increases the difficulty of fusion. Therefore, most existing fusion methods primarily focus on lower magnification factors, such as $4\times$, while relatively less attention is paid to higher magnifications, such as $64\times$. As hyperspectral imaging technology advances, there is an increasing demand for high-magnification fusion techniques. For instance, the AHSI hyperspectral camera onboard the GF-5 satellite is equipped with 330 spectral bands covering the range from 400 to 2500 nm, with a spectral resolution of 5–10 nm and a spatial

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resolution of 30 m. In contrast, the WorldView-3 satellite is outfitted with a multispectral camera providing a panchromatic spatial resolution of 0.31 m (one panchromatic band) and a multispectral resolution of 1.24 m (eight multispectral bands). The difference in spatial resolutions between these two imaging technologies can reach 96×. Developing high-magnification fusion algorithms that preserve the advantages of both HSI and MSI is essential. This article addresses the challenges of 64× high-magnification image fusion.

In addition, the number of hyperparameters in a fusion algorithm plays a crucial role in its performance. As the number of hyperparameters increases, the complexity of tuning them rises, with even slight adjustments potentially leading to significant changes in fusion quality. Most existing fusion algorithms depend on multiple hyperparameters, but, in practical scenarios, fine-tuning them becomes difficult due to the absence of reference HR-HSI data. Consequently, eliminating the need for hyperparameters can help avoid the negative effects of poor parameter selection on fusion performance. Furthermore, there is an increasing demand for faster HSI and MSI fusion to support real-time decision-making more effectively [11]. For example, in the defense sector [12], real-time target detection and situational awareness are crucial for strategic decision-making. As a result, achieving high real-time performance in image fusion has become essential for the successful deployment of this technology.

To address the aforementioned issues, this article proposes a semiblind high-magnification fusion algorithm. This algorithm utilizes simple matrix operations and a specially designed spatial–spectral error backpropagation algorithm to maximize the retention of spectral information from the LR-HSI and spatial information from the HR-MSI. This approach significantly enhances the quality of high-magnification fusion, eliminates the need for hyperparameters, improves fusion speed, and resolves the semiblind fusion challenge by modifying the downsampling method. The method first uses the Moore–Penrose inverse of the SRF to obtain an initial fusion solution and then performs two stages to form the final fused HR-HSI image.

Spectral Optimization: The difference between the reconstructed HSI and the true HSI is treated as the spectral gradient error. A fusion equation is formulated based on this error and solved, updating the initial fusion solution. This step refines the fusion process with straightforward matrix multiplication and addition operations.

Spatial Optimization: The algorithm proceeds by reconstructing an MSI based on the fusion result from the previous step. The error between the reconstructed and original MSIs is then computed. This error, along with the pseudoinverse of the SRF, is used to generate an error image, which is added to the previous solution to produce the final precise fused image. Overall, the main innovations of this article are in three aspects.

1) This article introduces a novel algorithm that effectively preserves both spatial and spectral information by utilizing spatial–spectral errors, leading to a substantial improvement in the quality of fused images, especially for high-magnification difference images.
2) A novel semiblind fusion scheme is introduced, which replaces blurred downsampling with simple downsampling during the spectral error extraction phase, thereby eliminating the need for the PSF.
3) By utilizing comprehensive mathematical modeling, the method eliminates the need for hyperparameter settings, effectively preventing any negative impact of hyperparameter selection on fusion performance.

The remainder of this article is organized as follows. Section II introduces the related work. Section III provides a detailed description of the proposed method. Section IV outlines the experiments and analyzes various experimental results in detail. Section V presents the conclusions.

II. Related Work

Researchers have developed various fusion methods to generate images with both high spectral and spatial resolutions. These methods can be broadly categorized into two types: traditional fusion methods and deep learning-based fusion methods. Most traditional methods are nonblind, with some being semiblind, whereas some deep learning-based methods are blind.

A. Traditional Fusion Methods

Traditional methods for image fusion focus on enhancing spatial and spectral resolution through various approaches, such as detail injection, Bayesian estimation, matrix decomposition, and tensor representation. Detail injection-based methods, including component substitution (CS) [13] and multiresolution analysis (MRA) [14], work by first separating spatial and spectral information in low-resolution images. The CS method injects spatial details from high-resolution images back into the original image space for improved clarity, while MRA decomposes the input image into multiple resolutions, extracting high-frequency details absent in lower resolutions to enhance overall resolution. Classical implementations of detail injection include IHS transformation [15], Gram–Schmidt [16], Brovey [17], and wavelet transformations [18], each balancing spectral and spatial fusion but facing potential issues such as distortion and high computational complexity. Bayesian estimation-based methods apply prior information to regulate the target scene, aiming to solve an inverse problem by maximizing the posterior probability or minimizing the log-posterior probability, often utilizing subspace techniques. For instance, stochastic mixture models (MAP-SMMs) [19], [20] have been applied successfully to MSI and HSI image fusion, while models incorporating Gaussian mixture scale priors in the wavelet domain [21], [22] enhance noise robustness. However, Bayesian methods introduce variability from random number dependencies, resulting in occasional instability. Matrix decomposition methods simplify the data by transforming 3-D hyperspectral data into a 2-D matrix, assuming decomposability into two core matrices—endmembers and abundances [23], [24], [25] or sparse dictionary and coefficient representations [26]—which helps efficiently combine low-resolution hyperspectral data and high-resolution multispectral data. Finally, tensor

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representation-based methods view data as a third-order tensor, maintaining the original arrangement within the cube structure and decomposing data into low-dimensional components for more intuitive analysis. Techniques such as Tucker decomposition [27], CP decomposition [28], BDT decomposition [29], tensor networks [30], and t-SVD [31] perform well with complex data structures but present high computational demands, especially for large datasets or high ranks, and are sensitive to data noise and randomness.

B. Deep Learning-Based Fusion Methods

In recent years, numerous deep learning-based hyperspectral MSI fusion methods [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43] have been proposed. These methods are categorized into supervised and unsupervised approaches based on their reliance on labeled data. Supervised methods typically require HR-HSI as training labels to learn the mapping relationships between pairs or triplets composed of LR-HSI and HR-MSI with HR-HSI, thereby achieving super-resolution reconstruction. For example, Dian et al. [44] introduced a deep HSI sharpening method that enhances spatial resolution using convolutional neural networks (CNNs). Tian et al. [38] proposed an interpretable model-driven deep network optimized by incorporating deep priors and the proximal gradient descent (PGD) algorithm. Similarly, Dong et al. [34] presented a scale-driven diffusion model approach that integrates denoising diffusion probabilistic models (DDPMs) and the physical mechanisms of HSI-SR to achieve high-resolution reconstruction of HSI. Despite the clear advantage of supervised methods in terms of fusion accuracy, their application is constrained by the scarcity of HR-HSI data.

To address this limitation, researchers have proposed unsupervised deep learning methods that achieve super-resolution by introducing deep priors, physical model information, or leveraging the intrinsic characteristics of input images. For instance, Cao et al. [35] introduced an unsupervised hybrid network that combines Transformers and CNNs, fully exploiting the global similarity between multilevel features through a multilevel cross-feature attention mechanism to reconstruct high-resolution HSI. Dong et al. [37] employ a context-aware guided attention module to guide the feedback refinement of low-level features with high-level features, thereby enhancing the transfer of spatial–spectral features and achieving fusion. Qu et al. [36] refine registration and fusion results through iterative co-optimization of registration fusion. Gao et al. [43] proposed an enhanced autoencoder network (EU2ADL) that achieves unsupervised HSI super-resolution through degradation learning embedded with attention mechanisms. In addition, some deep learning methods also incorporate traditional model priors to enhance fusion effects. For example, Liu et al. [41] proposed an autoencoder inspired by traditional models, capable of effectively reconstructing high-resolution HSIs (HR-HSIs). Li et al. [45] combined deep image priors (DIPs) with traditional physical degradation model information to propose a multistage unsupervised network (M2U-Net), which realizes HSI fusion super-resolution (HISR). Although deep learning-based methods benefit from the advantages of network learning, they typically require substantial amounts of data for training simulation. In high-magnification fusion scenarios, where LR-HSI data are even more scarce, deep learning-based methods often exhibit suboptimal performance.

III. Proposed Method

This section will focus on describing our proposed semiblind high-magnification fusion framework, a rapid fusion method suitable for applications with high real-time requirements. Our core method utilizes a spatial–spectral error optimization approach, as shown in Fig. 1, which includes a pre-estimation phase and two error optimization stages. In the pre-estimation phase, MSI is used to estimate the fusion result quickly. In the spectral error optimization stage, the spectral gradient error is calculated and used to formulate a fusion equation. Solving this equation produces the spectral fusion error, which is then backpropagated. In the spatial error optimization stage, spatial errors from the multispectral matrix reconstruction are calculated using the M-P inverse of the SRF. These spatial errors are backpropagated, completing the second optimization and yielding the final fused HR-HSI.

A. Problem Formulation

In this article, we define the matrix of the target HR-HSI as $\mathbf{Z} \in \mathbb{R}^{L\times WN}$, where L represents the number of bands and $W \times N$ represents the number of pixels. Similarly, the matrix of the LR-HSI used for fusion is defined as $\mathbf{H} \in \mathbb{R}^{L\times wn}$, indicating that it contains L bands and $w \times n$ pixels. The HR-MSI involved in the fusion is defined as $\mathbf{M} \in \mathbb{R}^{l\times WN}$, where l represents the number of bands and $W \times N$ represents the number of pixels. According to the characteristics of HR-HSI, LR-HSI, and HR-MSI, it can be seen that $L > l, W > w$, and $N > n$. In matrix decomposition-based fusion models, it is generally assumed that LR-HSI is the spatially blurred and downsampled matrix of the target HR-HSI; therefore, (1) and (2) can be obtained, where $\mathbf{B} \in \mathbb{R}^{WN\times WN}$, $\mathbf{S} \in \mathbb{R}^{WN\times wn}$, and $\mathbf{R} \in \mathbb{R}^{l\times L}$ represent the spatial blur matrix, the downsampling matrix, and the SRF matrix, respectively. Since the hardware parameters of the camera determine the SRF, we assume that the SRF is known; however, the PSF is usually related to the shooting scene and camera, making it difficult to provide in practical applications, so it is set as an unknown parameter. This article addresses the problem of solving for matrix $\mathbf{Z}$ without using PSF through pre-estimate and spectral error optimization steps. The following is the pre-estimate part:

$$\mathbf{H} = \mathbf{Z} \cdot \mathbf{B} \cdot \mathbf{S} \tag{1}$$

$$\mathbf{M} = \mathbf{R} \cdot \mathbf{Z}. \tag{2}$$

To solve the fused matrix $\mathbf{Z}$, the majority of existing traditional methods, such as those based on matrix factorization [46] and those based on sparse decomposition [47], require the decomposition of matrix $\mathbf{Z}$ into the product of a spectral basis and a coefficient matrix

$$\mathbf{Z} = \mathbf{D} \cdot \mathbf{C} \tag{3}$$

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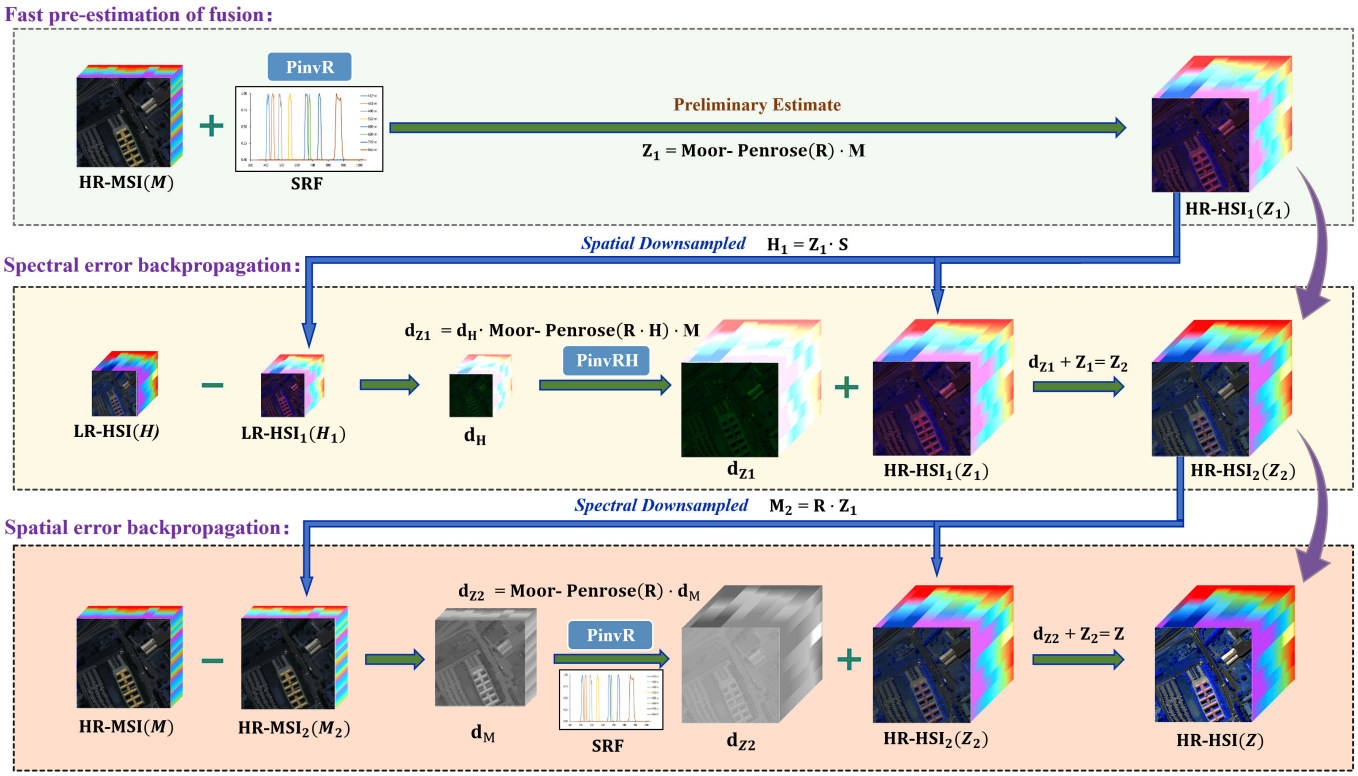


Fig. 1. Flowchart of the parameter-free spectral–spatial optimization algorithm for semiblind hyperspectral and MSI fusion: 1) fast pre-estimation of fusion: quickly estimate the HR-HSI solution Z_1 by using the Moore–Penrose inverse of the RSF and the product of the MSI; 2) spectral error backpropagation: perform spatial downsampling on Z_1, compute the spectral error by comparing it with the HSI, and backpropagate the error to Z_1 to obtain the intermediate result Z_2; and 3) spatial error backpropagation: apply spectral downsampling to Z_2, calculate the spatial error by comparing it with the MSI, and backpropagate the error to Z_2 to get the final solution Z.

where $\mathbf{D} \in \mathbb{R}^{L \times L_s}$ represents the spectral basis and $\mathbf{C} \in \mathbb{R}^{L_s \times WN}$ denotes the coefficient matrix. The problem of solving for matrix $\mathbf{Z}$ is, thereby, transformed into the problem of solving for the spectral basis and the coefficient matrix. Typically, these methods require the construction of the following ill-posed convex optimization problem to solve for matrices $\mathbf{D}$ and $\mathbf{C}$:

$$\min_{\mathbf{D},\mathbf{C}} \|\mathbf{H}-\mathbf{D}\cdot\mathbf{C}\cdot\mathbf{B}\cdot\mathbf{S}\|_F^2+\|\mathbf{M}-\mathbf{R}\cdot\mathbf{D}\cdot\mathbf{C}\|_F^2+\lambda\|\mathbf{D}\|+\mu\|\mathbf{C}\| \tag{4}$$

where $\lambda|\mathbf{D}| + \mu|\mathbf{C}|$ denotes the constraints on the solutions for the spectral basis and coefficient matrix. To solve (4), a complex iterative optimization process is typically required, such as employing the alternating direction method of multipliers (ADMMs) optimization method. This often demands substantial computational resources, and the introduction of constraints during the solution process can also lead to the introduction of manual hyperparameters. Consequently, the practicality of such methods is limited.

On the other hand, most existing deep learning-based methods [48] typically construct a loss function using (5). They leverage the neural network's capability to learn features to train the network to understand the mapping relationship for fusing images, thereby solving for the fused image $\mathbf{Z}$. This process also requires a substantial amount of data-driven learning and involves manual hyperparameters such as learning rates and training epochs, which similarly impose limitations on the practicality of these methods

$$f(\mathbf{Z}) = \frac{1}{2}\|\mathbf{H}-\mathbf{Z}\cdot\mathbf{B}\cdot\mathbf{S}\|_F^2 + \frac{1}{2}\|\mathbf{M}-\mathbf{R}\cdot\mathbf{Z}\|_F^2. \tag{5}$$

This article, aiming to overcome these limitations, employs a nondecomposition method that directly solves for matrix $\mathbf{Z}$ by propagating spatial–spectral errors. Long et al. [7] solve the matrix $\mathbf{Z}$ by using prior information on singular values, whereas we simplify that method. Specifically, as shown in (3), we first obtain a preliminary estimate of the HR-HSI matrix $\mathbf{Z}_1$ directly using the Moore–Penrose inverse of the SRF and the HR-MSI matrix. This step provides an initial approach to addressing the challenge of insufficient spatial information in high-magnification super-resolution image fusion. Then we perform two additional optimization steps on this preliminary fused HR-HSI, progressively enhancing the spectral details from the HSI and the spatial details from the MSI, further refining and enriching the final fusion outcome

$$\mathbf{Z}_1 = \text{Moore-Penrose}(\mathbf{R})\cdot\mathbf{M}. \tag{6}$$

B. Spectral Error Optimization

In this step, we formulated the spectral error calculation equation and replaced the blurred downsampling method with direct downsampling. This adjustment facilitated the extraction and feedback of spectral information, thereby enhancing the richness of spectral details in the fusion results. Furthermore, it successfully overcame the challenge of avoiding the use of the PSF function.

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Equation (6) can be used to quickly estimate a rough solution for HR-HSI using the SRF and MSIs. However, since (6) does not utilize LR-HSI, the quick solution $\mathbf{Z}_1$ for HR-HSI lacks spectral information, resulting in suboptimal reconstruction quality. Therefore, introducing spectral information into the quick solution $\mathbf{Z}_1$ has become an important direction for improving the reconstruction quality of HR-HSI.

One of the most direct methods to introduce spectral information is to reconstruct the LR-HSI image $\mathbf{H}_1$ through $\mathbf{Z}_1$ and then calculate the error $\mathbf{d}_H$ between the original LR-HSI and the reconstructed image $\mathbf{H}_1$ obtained by spatial downsampling using $\mathbf{Z}_1$. At this point, $\mathbf{d}_H$ contains most of the spectral error information between the quick solution $\mathbf{Z}_1$ and the target HR-HSI. The following describes the process of obtaining $\mathbf{d}_H$:

$$\mathbf{H}_1 = \mathbf{Z}_1 \cdot \mathbf{B} \cdot \mathbf{S} \tag{7}$$

$$\mathbf{d}_H = \mathbf{H} - \mathbf{H}_1. \tag{8}$$

Note that in the aforementioned (7), the fuzzy matrix $\mathbf{B}$ is used. Since the method proposed in this article is a semiblind fusion method, where the point spread function (PSF) is unknown, we use (9) to replace (7) to maintain generality. It should be noted that subsequent experiments demonstrate that this replacement does not affect the fusion performance

$$\mathbf{H}_1 = \mathbf{Z}_1 \cdot \mathbf{S}. \tag{9}$$

In this step, as we use the preliminary fused matrix to reconstruct the simulated HR-HSI and the matrix is inherently blurred, the difference between direct downsampling and blurred downsampling becomes negligible. Therefore, we opt for direct downsampling, bypassing the need for blurred downsampling. This strategy helps overcome the fusion method's reliance on the PSF, allowing our approach to achieve semiblind fusion.

$\mathbf{d}_H$ obtained in (8) contains most of the spectral error information. To utilize this spectral information to enhance $\mathbf{Z}_1$ and obtain an HR-HSI estimate rich in spectral information, we introduce an error variable $\mathbf{d}_{Z1}$ with the same dimensions as $\mathbf{Z}_1$ to retain these spectral error details. Here, $\mathbf{d}_{Z1}$ represents the error between our fast estimate $\mathbf{Z}_1$ of the HR-HSI and the actual value $\mathbf{Z}$ of the HR-HSI, i.e., $\mathbf{d}_{Z1} = \mathbf{Z} - \mathbf{Z}_1$ in the ideal case. However, since the actual value $\mathbf{Z}$ of the HR-HSI is unknown, $\mathbf{d}_{Z1}$ must be solved based on its physical model relationship with $\mathbf{d}_H$. Therefore, it is necessary to establish the relationship between $\mathbf{d}_H$ and $\mathbf{d}_{Z1}$ so that the spectral information in $\mathbf{d}_H$ can be added to $\mathbf{Z}_1$ through $\mathbf{d}_{Z1}$. Inspired by (1), we can assume that there is a relationship between $\mathbf{d}_H$ and $\mathbf{d}_{Z1}$, as shown in the following equation:

$$\mathbf{d}_H = \mathbf{d}_{Z1} \cdot \mathbf{B} \cdot \mathbf{S}. \tag{10}$$

Thus, solving the spectral error is converted into solving for $\mathbf{d}_Z$ from (10). The most direct way to solve $\mathbf{d}_Z$ can be through (11). The notation $()^{-1}$ represents the matrix inverse operation

$$\mathbf{d}_{Z1} = \mathbf{d}_H \cdot (\mathbf{B} \cdot \mathbf{S})^{-1}. \tag{11}$$

Equation (11) involves solving for the inverse of the product matrix $\mathbf{B}\cdot\mathbf{S}$ matrix. However, the dimensions of matrices $\mathbf{B}$ and $\mathbf{S}$ are usually large, and directly computing $(\mathbf{B} \cdot \mathbf{S})^{-1}$ requires substantial computational resources and memory. Particularly, in high-magnification fusion tasks, this computational burden may lead to inefficient algorithms and even render them inapplicable in real-world scenarios. Therefore, it is necessary to consider other methods for substitution. From (1), we know that the LR-HSI matrix $\mathbf{H}$ can be expressed as $\mathbf{Z} \cdot \mathbf{B} \cdot \mathbf{S}$. Consequently, we derive the following equation:

$$\mathbf{B} \cdot \mathbf{S} = \mathbf{Z}^{-1} \cdot \mathbf{H}. \tag{12}$$

At this point, (11) can be expressed as

$$\mathbf{d}_{Z1} = \mathbf{d}_H \cdot (\mathbf{Z}^{-1} \cdot \mathbf{H})^{-1}. \tag{13}$$

Equation (13) still contains an unknown matrix $\mathbf{Z}$. Since (2) shows $\mathbf{M} = \mathbf{R} \cdot \mathbf{Z}$, we consider introducing $\mathbf{M}$ to solve the problem of the unknown matrix $\mathbf{Z}$

$$\mathbf{d}_{Z1} = \mathbf{d}_H \cdot \left(\left(\mathbf{R}^{-1} \cdot \mathbf{M} \right)^{-1} \cdot \mathbf{H} \right)^{-1}. \tag{14}$$

In (14), all the matrices are known. However, since the equation contains multiple inverse operations $()^{-1}$, and matrices do not guarantee the requirement of being full rank, the inverse operations may introduce errors, and the accumulation of multiple inverse operations may amplify these errors. Therefore, we make a simple combination of matrix inverses

$$\mathbf{d}_{Z1} = \mathbf{d}_H \cdot (\mathbf{R} \cdot \mathbf{H})^{-1} \cdot \mathbf{M}. \tag{15}$$

Equation (15) contains only one inverse operation. Then, we use the Moore–Penrose inverse to replace the $()^{-1}$ operation to obtain the following equation:

$$\mathbf{d}_{Z1} = \mathbf{d}_H \cdot \text{Moore-Penrose}(\mathbf{R} \cdot \mathbf{H}) \cdot \mathbf{M}. \tag{16}$$

Thus, $\mathbf{d}_Z$ contains all the spectral errors between the $\mathbf{Z}_1$ matrix and the target HR-HSI, so a more accurate estimated value of the target HR-HSI can be obtained through the following equation:

$$\mathbf{Z}_2 = \mathbf{Z}_1 + \mathbf{d}_{Z1}. \tag{17}$$

C. Spatial Error Optimization

The more precise solution for $\mathbf{Z}_2$ obtained through (17) still contains spatial information errors in the newly introduced spectral information within $\mathbf{Z}_2$. To further enhance the results, we apply spatial error optimization to improve the fusion performance.

Similarly, we can perform spectral downsampling on the $\mathbf{Z}_2$ matrix to obtain the reconstructed multispectral matrix $\mathbf{M}_2$ and then subtract the reconstructed multispectral matrix from the original multispectral matrix $\mathbf{M}$ to obtain the error multispectral matrix $\mathbf{d}_M$. This process can be described by the following equations:

$$\mathbf{M}_2 = \mathbf{R} \cdot \mathbf{Z}_2 \tag{18}$$

$$\mathbf{d}_M = \mathbf{M} - \mathbf{M}_2. \tag{19}$$

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Similar to $\mathbf{d}_H$, we also consider transmitting the spatial information contained in $\mathbf{d}_M$ to $\mathbf{Z}_2$ through $\mathbf{d}_{Z2}$. Inspired by (2), we can obtain the following equation:

$$\mathbf{d}_M = \mathbf{R} \cdot \mathbf{d}_{Z2}. \tag{20}$$

Therefore, $\mathbf{d}_{Z2}$ can be determined by (18)

$$\mathbf{d}_{Z2} = (\mathbf{R})^{-1} \cdot \mathbf{d}_M = \text{Moore-Penrose}(\mathbf{R}) \cdot \mathbf{d}_M. \tag{21}$$

We can utilize the $\mathbf{d}_Z$ obtained from the spatial error through (22) as further optimization information to enhance the fusion performance

$$\mathbf{Z} = \mathbf{Z}_2 + \mathbf{d}_{Z2}. \tag{22}$$

The above section offers a detailed explanation of our proposed algorithm, which is fully based on matrix operations and eliminates the need for hyperparameters to control fusion quality. This approach effectively bypasses the challenges associated with hyperparameter tuning. Moreover, the algorithm is streamlined, significantly improving the fusion speed and resolving the issue of slow fusion.

Algorithm 1 Algorithm

Require: $\mathbf{H}, \mathbf{M}, \mathbf{R}$
  Get $\mathbf{Z}_1$ by equation (6)
  Spectral Error Optimization:
    Get $\mathbf{H}_1$ by equation (9)
    Get $\mathbf{d}_H$ by equation (10)
    Get $\mathbf{d}_{Z1}$ by equation (16)
    Get $\mathbf{Z}_2$ by equation (17)
  Spatial Error Optimization:
    Get $\mathbf{M}_2$ by equation (18)
    Get $\mathbf{d}_M$ by equation (20)
    Get $\mathbf{d}_{Z2}$ by equation (21)
    Get $\mathbf{Z}$ by equation (22)
  return Final Solution: $\mathbf{Z}$

IV. Experiments

To validate the effectiveness of our proposed method, we compared it with 12 state-of-the-art algorithms. These comparison algorithms include six traditional fusion methods, OTD [49], NFF [7], LTMR [46], RLR-MDF [50], LS-MDF [50], and CTDF [51], and six deep learning-based methods, CUCaNet [52], MIAE [41], EU2ADL [43], SSR-NET [40], MHF-net [42], and DCFormer [53]. It is worth noting that in the comparative experiments, only the NFF method and our proposed method require only the SRF for simulation, whereas other methods necessitate both the PSF and SRF. Before the introduction of our method, NFF was the best-performing semiblind fusion algorithm among the compared methods. Furthermore, to demonstrate that semiblind fusion is more challenging than nonblind fusion, we compared the nonblind and semiblind versions of the MIAE and EU2ADL algorithms in our experiments. For the nonblind fusion versions, the true PSF and SRF were provided during the fusion process. For the semiblind fusion version of MIAE, the PSF was estimated by its built-in blind estimation subnetwork, and the fusion result was obtained in combination with the true SRF. For the semiblind fusion version of EU2ADL, the PSF was estimated by the blind estimation subnetwork from another algorithm, M2U-Net [45], proposed by the same authors, and the fusion result was obtained in combination with the true SRF. It should be noted that M2U-Net [45] was not directly used for comparison because it is incapable of performing fusion at high magnifications such as 64×. The source codes for all the comparison methods are publicly available. Since deep learning methods require data training using a GPU, these six deep learning methods were run on a server with good hardware conditions and were equipped with four NVIDIA TITAN RTX GPUs. All other methods were run on a local host with an Intel[1] Core i7-11700 CPU and 16-GB DDR memory.

A. Dataset

In this article, we used two simulated datasets and three real datasets. The simulated datasets were sourced from the Pavia University and Harvard datasets. The real datasets were obtained from the Sentinel-2A satellite, the Earth Observing-1 satellite, and the ZY-1 02D satellite, as well as from images taken by our research group at a middle school playground in Tianjin, China.

The simulated datasets used in this article are given as follows.

1) *Pavia University Remote Sensing Dataset:* This dataset was captured by the ROSIS sensor over Pavia, a city in northern Italy. The original Pavia University dataset contains 115 bands with a resolution of 61 × 340 pixels. After excluding some undesirable bands, we selected a 256 × 256 × 93 pixel image from the top-left corner as the reference image.
2) *Harvard Ground-Based Dataset:* This dataset includes 50 images taken under daylight conditions both indoors and outdoors, as well as 27 images taken indoors under artificial or mixed lighting. Each image contains 31 bands with a resolution of 1040 × 1392 pixels. To ensure the generality of the results, we randomly selected one image and used a 512 × 512 × 31 pixel section from the top-left corner as the reference image.

The real datasets used in this article are given as follows.

1) *SE Dataset:* This dataset is composed of two parts. The MSIs are sourced from the Sentinel-2A satellite, containing 13 spectral bands with a ground spatial resolution of 10 m. We extracted 300 × 300 pixel regions from these images and selected four bands as the HR-MSI. The HSI is obtained from the Hyperion sensor on the Earth Observing-1 satellite, which provides images with 220 spectral bands and a ground resolution of 30 m. After removing the water absorption bands, we ultimately selected 100 × 100 pixel images containing 89 bands as the low-resolution LR-HSI.
2) *ZY Dataset:* It is from the ZY-1 02D satellite; the original data include an LRHSI with 166 bands and an HR-MSI with eight bands, with dimensions of

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TABLE I
ABLATION EXPERIMENT ON THE PAVIA UNIVERSITY DATASET

Pavia University Dataset						
Methods	**PSNR**	**RMSE**	**ERGAS**	**SAM**	**SSIM**	**TIME(S)**
Our(NSpatial-NSpectral)	27.979	19.782	0.632	21.208	0.853	0.027
Our(NSpatial-**Spectral**)	27.086	17.278	0.488	11.727	0.798	0.071
Our(**Spatial**-NSpectral)	27.979	19.782	0.632	21.208	0.853	0.06
Our(Spatial-Spectral)	**42.841**	**2.293**	**0.076**	**2.336**	**0.99**	**0.115**

TABLE II
ABLATION EXPERIMENT ON THE HARVARD DATASET

Harvard Dataset						
Methods	**PSNR**	**RMSE**	**ERGAS**	**SAM**	**SSIM**	**TIME(S)**
Our(NSpatial-NSpectral)	33.346	12.797	2.194	34.912	0.773	0.045
Our(NSpatial-**Spectral**)	44.725	2.21	0.487	3.442	0.959	0.09
Our(**Spatial**-NSpectral)	33.346	12.797	2.194	34.912	0.773	0.076
Our(Spatial-Spectral)	**53.009**	**0.681**	**0.247**	**1.754**	**0.993**	**0.146**

2064 × 2028 and 6192 × 6048 pixels, respectively. It is worth noting that due to the low SNR of some bands in the ZY dataset, we selected only the first 76 bands of the LR-HSI for the fusion experiments. In addition, due to computational memory constraints, we used only a 512 × 512 pixel section from the top-left corner of the LR-HSI image and a 1536 × 1536 pixel section of the HR-MSI image.

3) *TJS Dataset:* This dataset was captured by our research group at a middle school playground in Tianjin, using an SPECIM-FX17E hyperspectral camera and an RGB camera. The spectral range of the hyperspectral camera is 0.935–1.72 μm, with a spectral resolution of 3.5 nm. After registration and alignment processing, the LR-HSI used in this article has dimensions of 333 × 422 × 224, containing 224 spectral bands. The HR-MSI is an RGB image with dimensions of 999 × 1266 × 3. Considering the large data volume, we simplified the computation by selecting every fifth band from the LR-HSI, resulting in an LR-HSI image of 333 × 422 × 44 pixels. To validate the fusion effect on real datasets and perform subsequent classification metric evaluations, we labeled nine classes in the image. These nine classes are labeled as "white_cloth," "Lawn," "Green_shrubs," "Crimson_shrubs," "Red_wall," "Black_cloth," "Soil_ground," "Plastic_track," and "Sand_ground."

B. Implementation Details

We employed several commonly used image evaluation metrics to assess the quality of the fusion. These metrics include the peak signal-to-noise ratio (PSNR) [46], which measures the ratio of the maximum possible power of a signal to the power of the residual error between the fused image and the reference image. A higher PSNR value indicates better fusion quality. A smaller root mean square error (RMSE) [46] value indicates higher fusion quality. The relative dimensionless global error in synthesis (ERGAS) [54] describes the overall quality of the fused image, with a smaller ERGAS value indicating a better fusion effect. The spectral angle mapper (SAM) [26] is used to evaluate spectral distortion after fusion; a smaller SAM value indicates higher spectral fidelity and better fusion quality. The structural similarity index (SSIM) [55] evaluates the similarity between the fused image and the reference image in terms of luminance, contrast, and structure, with an SSIM value closer to 1 indicating better fusion quality. In addition, the time taken for the fusion process was also considered, with shorter processing time indicating higher fusion efficiency.

In this article, we also assessed the fusion effect by evaluating the classification performance of the fused images. To verify the effectiveness of the proposed method on real datasets, we tested three classification metrics: overall accuracy (OA), average accuracy (AA), and the kappa coefficient (Kappa).

C. Experimental Setting

We rigorously validated the effectiveness of the proposed method through strict experiments. In the Pavia University dataset, we applied Gaussian blur (with a mean of 2 and a size of 7 × 7) to the reference image and then downsampled it to generate the LR-HSI used for fusion. Simultaneously, we spectrally downsampled the reference image using an IKONOS-like spectral response filter to simulate the HR-MSI. In the Harvard dataset, we applied the same Gaussian blur and downsampling methods to generate the LR-HSI and used the SRF of the Nikon D700 camera to generate the HR-MSI. To ensure the fairness of experimental comparisons, we optimized the parameters of all comparison methods to achieve the best fusion results (it is noteworthy that our proposed method adapts without the need for parameter adjustments).

The primary focus of the proposed method is to address fusion challenges arising from high-magnification super-resolution. Therefore, we compared the performance of various methods under 64× super-resolution on the Pavia University and Harvard datasets, validating the effectiveness of our proposed method through objective evaluation metrics and subjective image display. In addition, to demonstrate that our method maintains good fusion performance even under low magnification super-resolution, we conducted fusion experiments at 4×-64× magnification on these two datasets.

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Finally, to verify the practicality of the proposed method, we conducted experimental comparisons on real datasets. To ensure a relatively objective evaluation, we performed a semisimulated fusion comparison experiment on the real ZY dataset. We used the LR-HSI from the ZY dataset as the reference HR-HSI, downsampled it by 64× to obtain the LR-HSI for fusion, and spatially downsampled the HR-MSI in the real dataset by 3× to use it as the HR-MSI for fusion. We then calculated the objective metrics for fusion using the error between the fused image and the original LR-HSI.

Although this process allows for the calculation of fusion objective metrics, the downsampling operation (semisimulation) on the real data may introduce some errors. To further validate the effectiveness of our proposed method on real datasets, we directly used the real LR-HSI and HR-MSI as inputs for fusion and displayed the fusion results to demonstrate the actual fusion performance. In addition, we conducted classification experiments on the TJS real datasets collected by our research group and validated the effectiveness of the proposed method using the experimental results.

In the experiments on real datasets, all comparison methods used the SRF estimation method proposed in reference [56] to estimate the SRF. For simplicity and fairness, all comparison methods that required the use of PSFs utilized the same PSF. These PSFs were generated by Gaussian blur with a size of 7×7 and a mean of 2.

The following sections present an in-depth analysis of results from ablation studies, comparative tests, and fusion experiments on both synthetic and real datasets, highlighting the effectiveness and superiority of our proposed method.

D. Ablation Study Results And Analysis

To validate the effectiveness of the two error optimization modules, we tested variant versions of the proposed algorithm by removing each optimization module. These variants include fusion without spatial–spectral error optimization, fusion with only spectral error optimization, fusion with only spatial error optimization, and fusion with both spectral and spatial optimization.

Tables I and II present the ablation study results of each key module. Specifically, fusion without spatial–spectral optimization refers to directly using $\mathbf{Z}_1$ as the final fusion result for objective evaluation. Fusion with only spectral error optimization refers to using the result of $\mathbf{Z}_2$ as the final output for objective evaluation. Fusion with only spatial error optimization involves directly using the fusion result $\mathbf{Z}_1$, then inputting it into the spatial error optimization process, and subsequently using the output $\mathbf{Z}$ as the final result. The full spectral–spatial optimization refers to using the result $\mathbf{Z}$ produced by following Algorithm 1 as the final output.

As shown in Tables I and II, the effect after spatial–spectral optimization is significantly higher than when only one of the optimizations is used. It is also evident that after spectral optimization, the SAM metric is substantially reduced, which strongly demonstrates the effectiveness of the proposed spectral optimization. The experimental results in Tables I and II fully validate the effectiveness of the proposed dual-stage optimization.

TABLE III
EXPERIMENTS USING DIFFERENT PSFS ON THE PAVIA UNIVERSITY DATASET

Pavia dataset						
PSF	**PSNR**	**RMSE**	**ERGAS**	**SAM**	**SSIM**	**TIME(S)**
gaussian,7,2	42.841	2.293	0.076	2.336	0.99	0.115
gaussian,3,2	42.904	2.262	0.074	2.269	0.99	0.1
gaussian,15,2	42.927	2.269	0.075	2.32	0.99	0.098
average	42.852	2.274	0.075	2.282	0.99	0.096
disk	43.114	2.207	0.072	2.292	0.991	0.092

TABLE IV
EXPERIMENTS USING DIFFERENT PSFS ON THE HARVARD DATASET

Harvard dataset						
PSF	**PSNR**	**RMSE**	**ERGAS**	**SAM**	**SSIM**	**TIME(S)**
gaussian,7,2	53.009	0.681	0.247	1.754	0.993	0.146
gaussian,3,2	53.015	0.679	0.248	1.753	0.993	0.116
gaussian,15,2	53.01	0.681	0.247	1.754	0.993	0.118
average	53.015	0.679	0.248	1.753	0.993	0.107
disk	53.01	0.681	0.247	1.755	0.993	0.105

TABLE V
NUMBER OF PARAMETERS THAT NEED TO BE TUNED FOR DIFFERENT METHODS

Method	**Number of required parameters**
LTMR	4
NFF	3
OTD	9
LS-MDF	2
RLR-MDF	4
CTDF	4
Ours	**0**

1) Applicable to Different PSFs: Tables III and IV show the fusion performance of our proposed method under 64× super-resolution on two simulated datasets when different PSFs are used to generate LR-HSI. As shown in Tables III and IV, even when different PSFs are used to generate LR-HSI, our proposed method still achieves relatively stable fusion performance. This demonstrates that our method is applicable to different PSFs and exhibits better adaptability.

2) Advantage of Parameter-Free Operation: The selection of hyperparameters significantly impacts the performance of fusion algorithms. Table V summarizes the number of tunable hyperparameters required by the model-based algorithms discussed in this article. As shown in the table, only our proposed method does not require manual hyperparameter tuning, while all other methods require adjusting at least two hyperparameters. The OTD method, for instance, requires a manual setting of up to nine parameters. The setting of these hyperparameters can greatly affect fusion performance.

Taking the LTMR method as an example, Table VI illustrates the performance of the Pavia dataset when using LTMR for fusion. We compare the performance of the optimal set of hyperparameters with scenarios where the value of the atomic number hyperparameter is increased or decreased by 1. As shown in the table, when the atomic number is increased by 1 from the optimal set, the fusion PSNR value decreases

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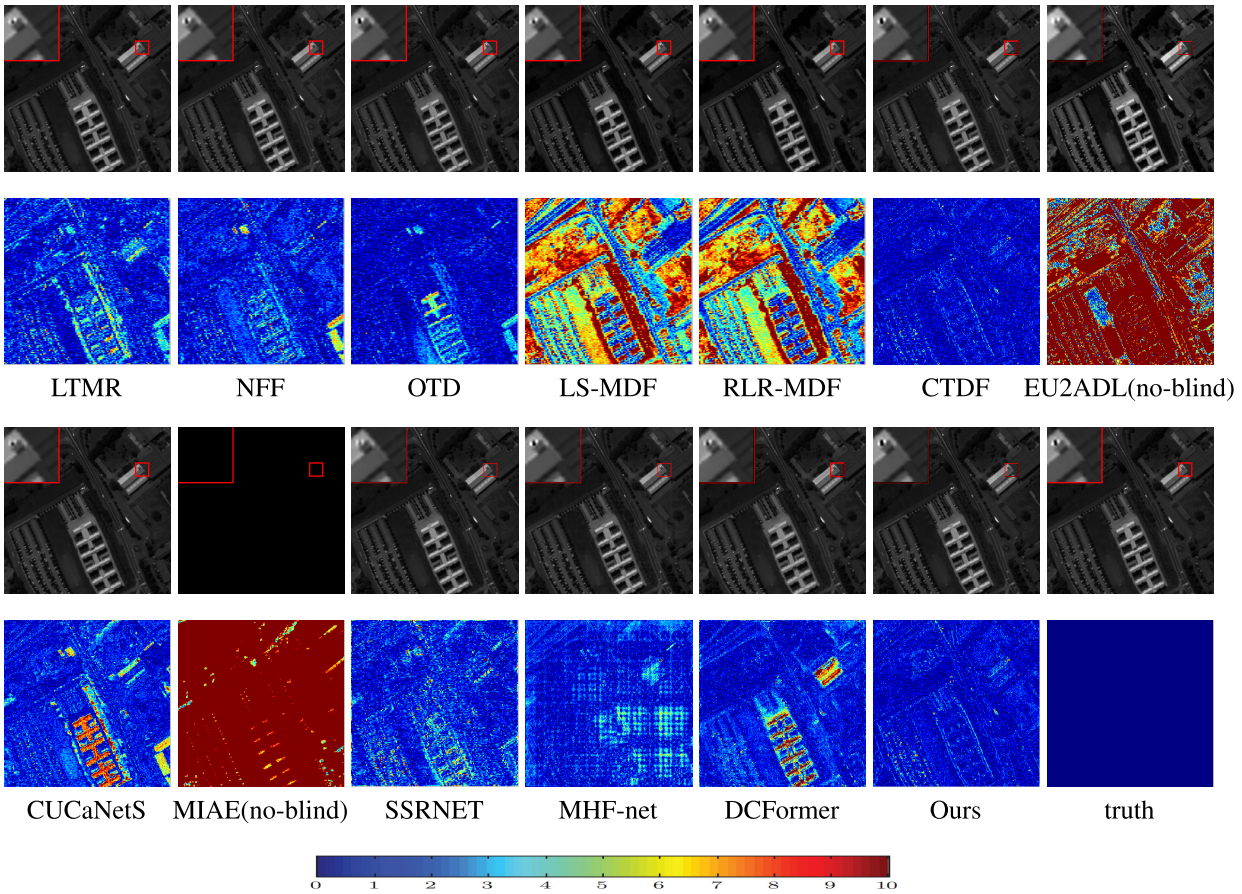


Fig. 2. Fused images of the 20th band of the various comparison methods on the Pavia University dataset. Odd-numbered rows show the fused images, and even-numbered rows show the corresponding error images, where the closer the error image is to the truth, the better.

TABLE VI
EXPERIMENT ON THE EFFECT OF ADJUSTING PARAMETERS ON FUSION RESULT

LTMR method on the Pavia Dataset						
	PSNR	RMSE	ERGAS	SAM	SSIM	TIME(S)
Best L	40.624	2.596	0.09	2.605	0.987	11.957
L+1	37.93	3.995	0.137	3.841	0.975	13.224
L-1	33.179	6.777	0.27	6.111	0.937	10.411

by nearly 3 dB, and when decreased by 1, the PSNR value drops by as much as 7 dB.

This demonstrates that adjustable hyperparameters have a substantial impact on fusion performance. On simulated datasets, hyperparameters can be manually tuned using ground-truth images of the fusion result. However, when applied to real datasets with no ground-truth images, it may be impossible to select the optimal hyperparameters, thereby affecting fusion performance. Moreover, in deep learning-based methods, the impact of hyperparameters is even more critical. These methods typically require manual tuning of at least two adjustable hyperparameters, such as learning rate and epochs. In addition, methods such as MIAE require the adjustment of two more parameters. Furthermore, deep learning-based methods necessitate adaptive adjustment of millions of hyperparameters during training, which is a major reason for the extensive training time required by these methods. Without appropriate hyperparameters, fusion performance is further compromised.

In contrast, our proposed method is entirely free of hyperparameter settings, thereby completely avoiding the negative impact of hyperparameter selection on fusion performance.

E. Fusion Performance at High Magnification

Table VII presents the fusion results of various methods on the Pavia University dataset at 64× magnification. The best performance among all metrics is indicated in **bold**, and the second-best performance is indicated with an underline. Notably, the SAM metric for the DCFormer method outperforms ours, while all other metrics are lower. This is due to DCFormer's use of HR-HSI training, which improves spectral vector alignment with the fusion results, leading to a higher SAM value. However, our method achieves optimal fusion in just 0.115 s, whereas the fastest alternative, LS-MDF, takes 0.253 s—over twice as long. Despite this longer processing time, LS-MDF's fusion performance is significantly inferior, underscoring the superior efficiency and effectiveness of our approach.

Fig. 2 presents the visual fusion results of all methods, with the image selected from the 20th band of the fused images. From the fused images in the first row, it can be seen that our method is very similar to the truth, making it difficult to distinguish any differences. The second row shows the error images for the 20th band compared to the truth; our method error image is almost entirely blue (the closer to blue, the

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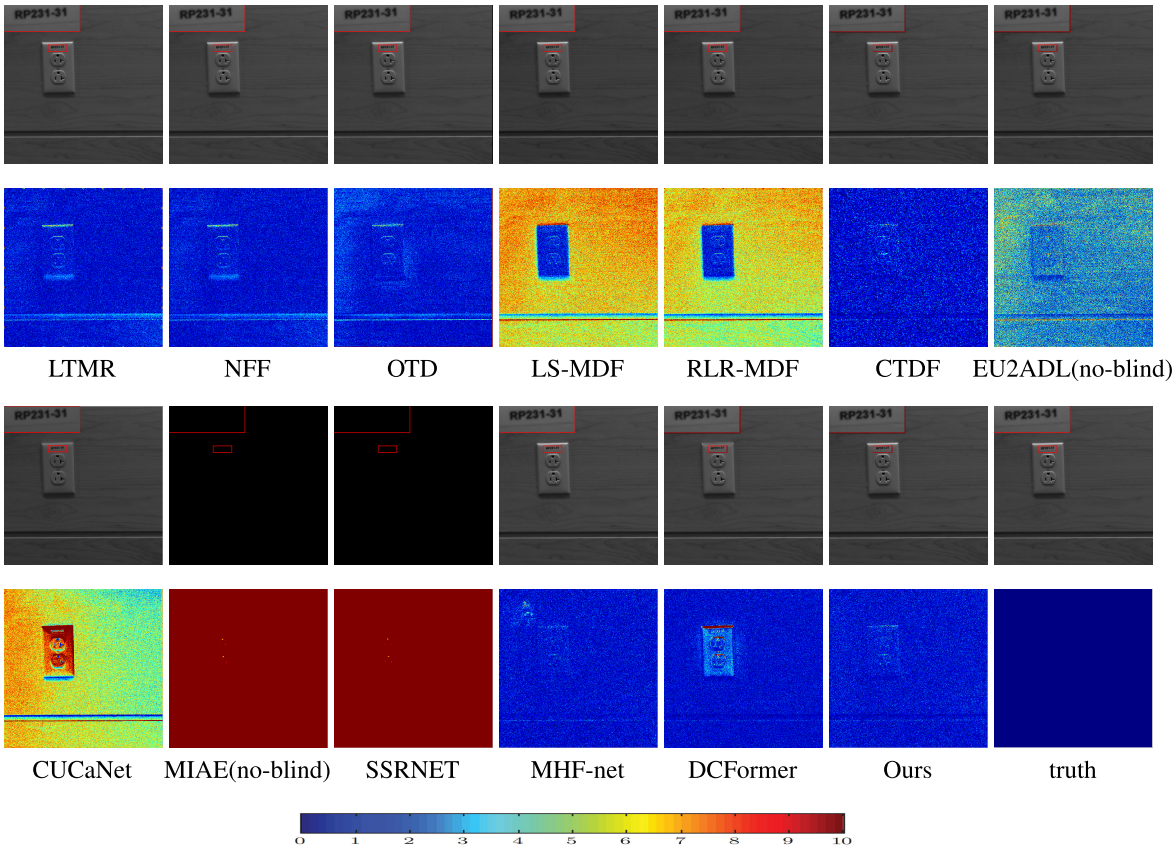


Fig. 3. Fused images of various comparison methods on the 20th band of the Harvard dataset. The first row displays the fused images, while the second row presents the corresponding error images, where the closer the error image is to the truth, the better.

TABLE VII

FUSION RESULTS OF FUSION METHODS ON THE PAVIA UNIVERSITY DATASET (64×)

Pavia University dataset(64X)								
Method	**Proposd Time and Source**	**PSNR**	**RMSE**	**ERGAS**	**SAM**	**SSIM**	**TIME(S)**	**Type**
LTMR	2019 (TIP) [46]	40.624	2.596	0.09	2.605	0.987	11.957	Traditional
NFF	2022 (IPT) [7]	40.232	2.863	0.1	2.898	0.988	1.252	Traditional
OTD	2020 (TIP) [49])	42.598	2.386	0.078	2.451	**0.99**	26.842	Traditional
LS-MDF	2020 (TIP) [50]	28.883	11.192	0.327	3.508	0.91	0.4808	Traditional
RLR-MDF	2020 (TIP) [50]	29.149	10.78	0.316	3.518	0.909	4.7045	Traditional
CTDF	2024 (TGRS) [51]	41.4131	3.4671	0.1005	3.1255	0.98	18.653	Traditional
CUCaNet	2020 (ECCV) [52]	38.256	3.763	0.988	3.79	0.984	2533	Deep Learning
MIAE(no-blind)	2022 (TGRS) [41]	13.692	55.459	14.63	89.864	0.006	358.56	Deep Learning
MIAE(semi-blind)	2022 (TGRS) [41]	13.692	55.459	14.63	89.864	0.006	384.937	Deep Learning
EU2ADL(no-blind)	2023 (TGRS) [43]	28.501	11.365	3.216	11.702	0.919	443.108	Deep Learning
EU2ADL(semi-blind)	2023 (TGRS) [43]	27.205	13.626	3.94	13.666	0.904	462.231	Deep Learning
SSRNET	2021 (TGRS) [40]	40.418	2.466	0.722	2.369	0.985	574.8	Deep Learning
MHF-net	2022 (TPAMI) [42]	35.965	4.905	1.4	6.004	0.975	6874.6	Deep Learning
DCFormer	2024 (IF) [53]	40.249	2.526	0.094	**1.817**	**0.99**	5146.3	Deep Learning
Ours	2024	**42.841**	**2.293**	**0.076**	2.336	**0.99**	**0.115**	Traditional

smaller the error with the truth; the closer to red, the larger the error, with the truth's error image being pure blue). Therefore, Fig. 2 clearly indicates that our method has the smallest fusion error.

Similarly, Table VIII displays the fusion results of all methods on the Harvard dataset, showing that under 64× magnification, our method again achieves state-of-the-art performance. The table also indicates that our method requires only 0.146 s, while the fastest alternative method, STEREO, takes 0.295 s—again, twice as long as our method. The fusion performance metrics of STEREO are also significantly lower than those of our method. In addition, Fig. 3 presents the visual fusion results of the various methods, demonstrating that the error map of our method's fusion results is closest to blue, further highlighting the effectiveness of our approach at high-magnification fusion.

Furthermore, in Tables VII and VIII, to highlight the greater difficulty of semiblind fusion, we also provided the fusion results of the nonblind and semiblind versions of the MIAE and EU2ADL algorithms. The results, especially those of the EU2ADL algorithm, show that the semiblind version slightly underperforms compared to the nonblind version. This is consistent with the experimental conclusions in the original literature [41], [45]. In addition, it can be observed from the

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TABLE VIII

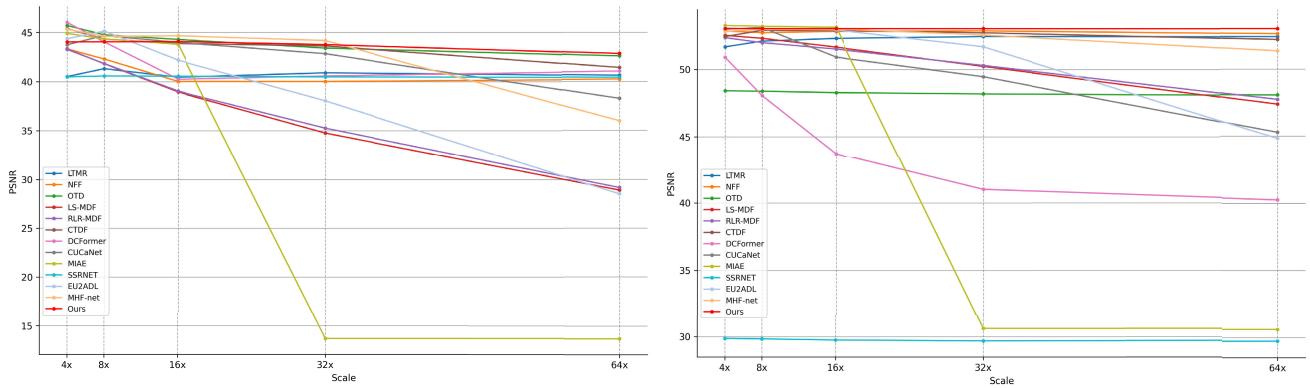
FUSION RESULTS OF FUSION METHODS ON THE HARVARD DATASET (64×)

Harvard dataset(64X)								
Method	**Proposd Time and Source**	**PSNR**	**RMSE**	**ERGAS**	**SAM**	**SSIM**	**TIME(S)**	**Type**
LTMR	2019 (TIP) [46]	52.416	0.722	0.265	1.828	0.992	4.696	Traditional
NFF	2022 (IPT) [7]	52.633	0.712	0.255	1.826	**0.993**	2.506	Traditional
OTD	2020 (TIP) [49]	48.067	2.018	0.287	2.558	0.992	62.826	Traditional
LS-MDF	2020 (TIP) [50]	47.382	2.257	0.31	1.887	0.987	0.295	Traditional
RLR-MDF	2020 (TIP) [50]	47.738	2.113	0.304	1.992	0.988	14.1329	Traditional
CTDF	2024 (TGRS) [51]	52.202	0.7193	0.252	1.77	0.992	20.623	Traditional
CUCaNet	2020 (ECCV) [52]	45.282	2.743	3.299	5.214	0.982	5039	Deep Learning
MIAE(no-blind)	2022 (TGRS) [41]	30.602	22.287	12.884	89.515	0.263	1661.7	Deep Learning
MIAE(semi-blind)	2022 (TGRS) [41]	30.602	22.287	12.884	89.515	0.263	1747.26	Deep Learning
EU2ADL(no-blind)	2023 (TGRS) [43]	44.822	6.607	6.204	16.766	0.949	1416.77	Deep Learning
EU2ADL(semi-blind)	2023 (TGRS) [43]	44.719	6.437	6.039	16.202	0.951	1424.29	Deep Learning
SSRNET	2021 (TGRS) [40]	29.729	21.15	61.567	66.05	0.52	957.9	Deep Learning
MHF-net	2022 (TPAMI) [42]	51.354	0.774	3.655	1.884	0.989	10965.77	Deep Learning
DCFormer	2024 (IF) [53]	49.508	1.504	0.272	1.769	0.991	78755.1	Deep Learning
Ours	2024	**53.009**	**0.681**	**0.247**	**1.754**	**0.993**	**0.146**	Traditional

TABLE IX

FUSION RESULTS OF FUSION METHODS ON THE ZY DATASET (64×)

ZY Dataset (64X)								
Method	**Proposd Time and Source**	**PSNR**	**RMSE**	**ERGAS**	**SAM**	**SSIM**	**TIME(S)**	**Type**
LTMR	2019 (TIP) [46]	31.496	10.161	0.176	4.151	0.712	27.248	Traditional
NFF	2022 (IPT) [7]	29.127	13.677	0.233	5.404	0.684	4.652	Traditional
OTD	2020 (TIP) [49]	28.548	13.815	0.238	3.461	0.969	153.348	Traditional
LS-MDF	2020 (TIP) [50]	29.357	12.953	0.223	3.882	0.692	1.9705	Traditional
RLR-MDF	2020 (TIP) [50]	28.53	14.107	0.243	3.968	0.678	21.7537	Traditional
CTDF	2024 (TGRS) [51]	28.8667	13.5216	0.2381	5.3342	0.6538	193.141	Traditional
CUCaNet	2020 (ECCV) [52]	17.04	45.944	1.212	11.007	0.592	2167	Deep Learning
MIAE(no-blind)	2022 (TGRS) [41]	12.106	91.889	1.58	89.804	0.01	3513.2	Deep Learning
EU2ADL(no-blind)	2023 (TGRS) [43]	28.494	19.183	6.066	11.02	0.686	2185.98	Deep Learning
SSRNET	2021 (TGRS) [40]	30.741	16.156	0.723	9.941	0.749	591.28	Deep Learning
MHF-net	2022 (TPAMI) [42]	12.106	91.889	1.58	89.804	0.01	2239	Deep Learning
DCFormer	2024 (IF) [53]	30.685	**9.195**	0.235	3.825	**0.817**	1575.3	Deep Learning
Ours	2024	**32.005**	10.137	**0.174**	**3.783**	0.728	**0.284**	Traditional



Pavia University Harvard

Fig. 4. PSNR value curve graph of various methods for 4×, 8×, 16×, 32×, and 64× fusion results on the Pavia and Harvard datasets.

tables that our proposed method, despite being a semiblind fusion approach, achieves better experimental results than other nonblind fusion methods. This further demonstrates the effectiveness of our proposed method.

F. Fusion Performance at Conventional Magnification

Fig. 4 shows the PSNR curves of various fusion methods on two simulated datasets at magnification factors of 4×, 8×, 16×, 32×, and 64×. As illustrated in the figure, although our proposed method is primarily designed for high-magnification fusion, it still maintains high fusion performance at conventional magnifications. Moreover, it can be seen that our method consistently achieves stable and high fusion performance across the magnification range from 4× to 64×. In contrast, most other methods experience a significant decline in fusion performance beyond 32× resolution. This demonstrates that our proposed method is highly robust to different magnification factors.

G. Fusion Performance on Real Datasets

1) Semisimulated Fusion on Real Datasets: In our experiments, we used the LR-HSI from the real dataset ZY

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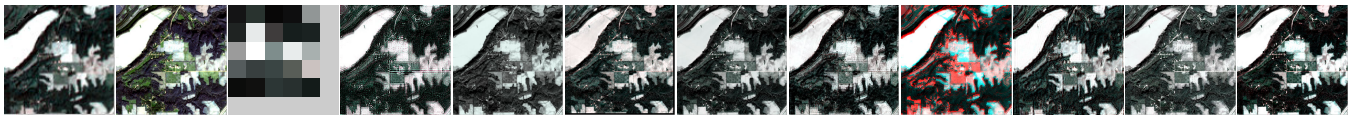


Fig. 5. Fused images of the various comparison methods on the SE real datasets. It shows the first, second, and third bands of the resulting fused image.

TABLE X
VISUAL DISPLAY OF CLASSIFICATION PERFORMANCE ON THE TJS REAL DATASETS

TJS Dataset								
Method	**LTMR**	**NFF**	**LS-MDF**	**RLR-MDF**	**MIAE**	**CTDF**	**Ours**	**Ours60X**
Kappa(%)	94.32	93.87	97.50	91.62	96.72	92.47	**97.55**	**97.63**
OA(%)	95.60	95.23	98.07	93.47	97.47	93.40	**98.10**	**98.17**
AA(%)	91.24	92.22	96.27	87.93	94.30	76.83	**96.32**	**96.41**

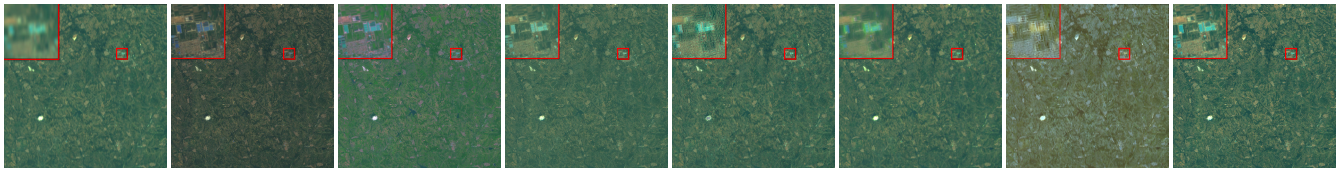


Fig. 6. Fused images of the various comparison methods on the ZY real datasets. It shows the first, second, and third bands of the resulting fused image.

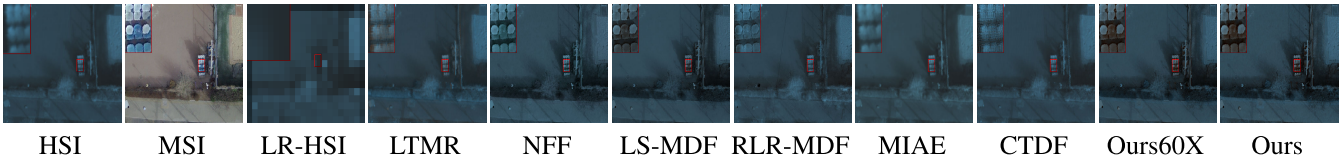


Fig. 7. Fused images of the various comparison methods on the TJS real datasets. It shows the first, second, and third bands of the resulting fused image.

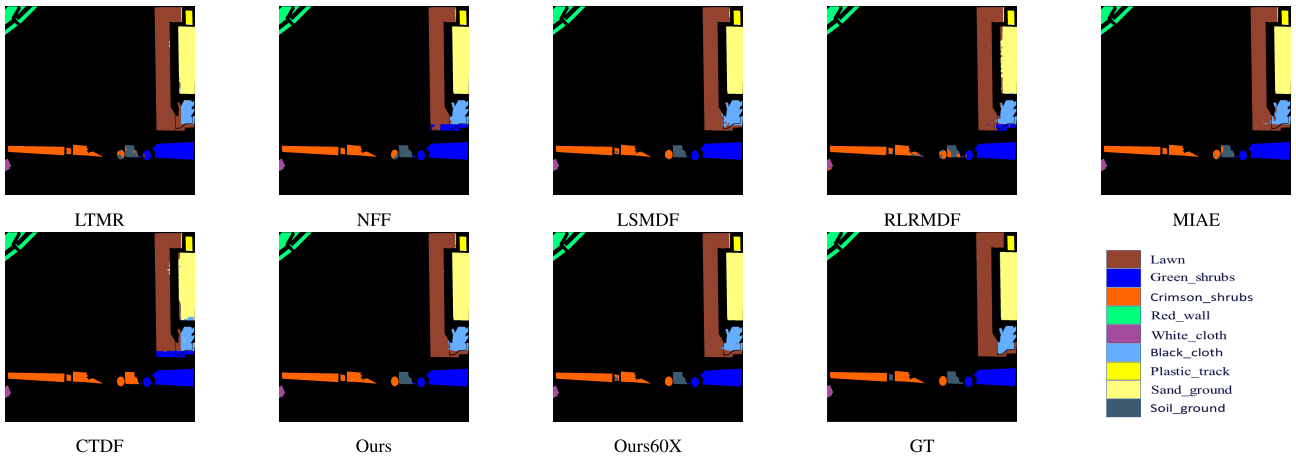


Fig. 8. Classification results of different fusion images using the FullyContNet classification algorithm.

as the reference HR-HSI. We then downsampled the LR-HSI by a factor of 64 to obtain an even low-resolution HSI, which was used as the LR-HSI in the semisimulated fusion. The HR-MSI from the real dataset was downsampled by a factor of 3 for use as the HR-MSI in the semisimulated fusion. Using the original LR-HSI as a reference, we calculated the metrics for the images produced by various fusion methods.

To enrich the comparative experimental results on the semisimulated dataset, during the training phase of the semisimulated experiment, we provided the reference HR-HSI as the training target for four supervised deep learning-based methods (SSRNET, CUCaNet, MHF-Net, and DCFormer). Table IX lists the semisimulated fusion results of different methods on the real dataset Y. The table shows that under 64× fusion, our proposed method has the shortest fusion time and the best overall fusion quality among all compared methods. In addition, the deep learning-based method DCFormer slightly outperformed our proposed method in two metrics. This is because such algorithms were provided with an additional reference HR-HSI as the training target, which endowed them with more prior information. However, despite this advantage, due to the limited available training data, these methods generally underperformed compared to our proposed method in high-magnification super-resolution scenarios. This

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indicates that our proposed method maintains strong fusion performance even on real datasets.

2) Visual Display of Fusion Results on Real Datasets: To further validate the usability of our proposed method on real datasets, we directly used the LR-HSI and HR-MSI from three datasets as inputs to generate the corresponding fused images. It is important to note that since HR-HSI cannot be provided to SSRNET, CUCaNet, MHF-Net, and DCFormer for network training on real datasets, the fusion results of these three methods are not included in this section.

Fig. 5 shows the fusion results of various methods on the first real dataset (SE dataset). From the visual display of the fused images, it can be observed that the images fused by our proposed method are clearer. In addition, the pseudocolor images of the fused results are very close to those of the LR-HSI, indicating that the images fused by our method have minimal spectral error. Fig. 6 shows the fusion results of the second real dataset (ZY dataset). It is worth noting that due to the high dimensionality of the ZY dataset, the NFF method fails to generate meaningful outputs, so its results are not displayed here. Similarly, it can be observed that the images fused by our proposed method possess the same level of clarity as the HR-MSI and exhibit pseudocolor effects similar to those of the LR-HSI. Likewise, Fig. 7 shows the fusion results on the third real dataset (TJS dataset), where it can also be observed that our proposed method produces superior fusion results.

3) Classification Performance of Fused Images on Real Datasets: To further verify the effectiveness of our proposed method on the TJS real datasets, we used the fused images from various algorithms as input for classification tasks and evaluated the fusion effectiveness using classification results. In this study, we used the publicly available FullyContNet [57] algorithm for classifying the fused images. The number of epochs in FullyContNet [57] was set to 100, and the other parameters were kept at their original settings. During the preprocessing phase, the PCA parameters for the fused images were set to 15. We used 1% of the data as the training set, 1% as the validation set, and 98% as the test set.

Table X shows the average values of the classification metrics, OA, AA, and Kappa, after five classification operations on the fused images produced by different fusion methods on our real datasets. The best results are highlighted in red, and the second-best results are highlighted in orange. The table indicates that the images fused by our proposed method achieved the best classification results. Therefore, considering both the subjective visual effects and the objective classification metrics, our proposed method demonstrates not only excellent visual quality but also outstanding classification performance. Notably, when achieving high-magnification super-resolution (Ours60×), our fusion method outperformed all other methods in both subjective visual effects and objective classification metrics. Fig. 8 shows the classification results using the FullyContNet [57] algorithm on the various fused images. The classification results further indicate that the images fused by our proposed method have better classification performance, confirming the practicality of our proposed method.

V. Conclusion

In this article, we propose a hyperspectral and MSI fusion algorithm that constructs a convex optimization function by utilizing spectral and spatial errors, effectively preserving both spectral and spatial information in the fused images and enabling high-magnification rapid fusion. Notably, the proposed method is a semiblind fusion approach that does not require the provision of PSF information during the fusion process, making it more suitable for a wide range of practical application scenarios. In addition, the method is entirely implemented using matrix calculations, employing a simple computational process to achieve high-quality, high-magnification super-resolution fusion. This approach avoids the complexities of traditional optimization processes, significantly speeding up the fusion and reducing processing time. Finally, fusion experiments were conducted on two simulated datasets and three real hyperspectral and multispectral datasets. The experimental results show that the proposed method does not require parameter settings and is applicable to any PSF function, delivering excellent fusion quality and speed across different magnification levels in super-resolution fusion, demonstrating strong practical applicability. Similarly, it can be seen that our proposed method still has significant room for improvement at lower magnification factors. In the future, more refined error backpropagation schemes can be explored based on the error backpropagation mechanism proposed in this article to further enhance the overall fusion performance of the algorithm.

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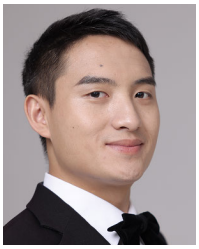
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